ARUMUGAM ABILASHAN

Multidisciplinary Automation & IoT Engineer

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Arumugam Abilashan

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PROFESSIONAL SUMMARY

Innovative Instrumentation and Automation Engineer with hands-on experience bridging industrial control systems (OT) with modern web and cloud technologies (IT). Proven track record in optimizing automated manufacturing lines in high-volume production environments. Highly skilled in PLC programming, mechanical integration, and developing custom IoT architectures using protocols like MQTT and web APIs. Adept at conceptualizing and deploying cross-disciplinary projects—from hardware data acquisition to interactive web interfaces—aimed at driving smart manufacturing, predictive maintenance, and operational efficiency.

WORK EXPERIENCE

HEMAS CONSUMER BRANDS, SRI LANKA : AUTOMATION ENGINEER INTERN ***FEB 2025 - SEP 2025***

• Supported industrial automation projects by designing and implementing automated quality control systems and mechanical packaging mechanisms.
• Integrated industrial motor controls, including VFDs and Servo drives, to optimize production workflows and equipment reliability.
• Applied Lean manufacturing concepts, 5S methodology, and Kaizen practices to minimize machine downtime and streamline factory floor operations.

KKR ROBOTICS (REMOTE) , INDIA : CAD DESIGNER TRAINEE ***FEB 2025- APR 2025***

• Designed 3D CAD models for educational robot prototypes utilizing SolidWorks and Autodesk Fusion 360.
• Developed detailed robotic body models and structural components to support learning-based hardware applications.
• Created comprehensive fabrication models to facilitate the accurate manufacturing and physical assembly of robotic parts.

SKILLS

- **Industrial Automation** - PLC Programming (Siemens, Mitsubishi, OpenPLC, Haiwell, Allen-Bradley), SCADA (Ignition), Advanced Instrumentation, Control System Modeling, Motor Control (Servo & Induction Motors), Servo & VFD configurations ,Pneumatic System Design, Power Electronics, Lean Manufacturing (5S, Kaizen), Factory Simulation
- **IoT & Embedded Systems** - MQTT, Modbus, Data Acquisition (DAQ), Web Bluetooth API, Raspberry Pi, ATmega328P, ESP32, Industrial Sensor Integration
- **AI & Software Development** -Machine Learning Applications, Computer Vision (OpenCV, MediaPipe), Data Analysis & Visualization, Python (NumPy, Pandas, Multiprocessing), React.js, Node.js, Three.js, C/C++
- **Electrical & Hardware Design** - SolidWorks, Autodesk Fusion 360, SketchUp, Mechanical Design, Simulation & Analysis (Ansys), Power System Modeling (Shielding & Lightning Protection), Panel Board Wiring & Installations, PCB Prototyping
- **Soft Skills** - Cross-functional Collaboration, Problem Solving, Project Management, Critical Thinking

SELECTED PROJECTS

View interactive demonstrations, 3D models, and full technical details for all projects at: [arumugam-abilashan.web.app]

1.POWER SYSTEM RELIABILITY MODELING FRAMEWORK | ***AUG 2025- PRESENT***

FINAL YEAR RESEARCH, UNIVERSITY OF COLOMBO

- Developed a novel framework utilizing Sequential Monte Carlo Simulations (SMCS) to quantify the reliability of renewable-rich island power systems, implementing multiprocessing and vectorized computing in Python for enhanced calculation efficiency.
- Evaluated complex system stability by analyzing reliability indices including LOLP, LOLE, and LOEE.

2. AUTOMATED QUALITY CONTROL (EMPTY CARTON BOX DETECTION - CLOGARD) | ***MAY 2025- JUN 2025***

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- Engineered a synchronized defect detection system utilizing a KD5022 capacitive sensor, Mitsubishi FX1N-60MT PLC, and encoder position tracking, programming the shift register logic via GX Developer.
- Designed the pneumatic rejection mechanism in SolidWorks and led its physical integration, successfully automating the process to increase line reliability and significantly reduce compressed air consumption.

3. PRODUCTION LINE OPTIMIZATION (KUMARIKA FEEDSCREW MECHANISM) | ***FEB 2025- JUL 2025***

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- Modeled a custom precision feedscrew mechanism using 3D CAD to synchronize and regulate bottle spacing for the Kumarika shampoo automated capping sequence.
- Facilitated the fabrication and physical installation of the feedscrew, eliminating mechanical bottlenecks, reducing line jams, and boosting the Overall Equipment Effectiveness (OEE).

4. END-OF-LINE AUTOMATION (PALLETIZING SYSTEM) | ***FEB 2025- SEP 2025***

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- Engineered a fully automated end-of-line palletizing workflow by integrating a custom YAVA conveyor system with an SZGH 4-DOF palletizing robot utilizing a vacuum-based end effector.
- Programmed the system's control architecture via a Haiwell PLC, successfully replacing manual material handling to substantially increase factory throughput and enhance overall workplace safety.

5. MECHANICAL AUTOMATION & CAD (SOAP GEAR CAM ROTARY PICK-AND-PLACE) | ***MAY 2025- SEP 2025***

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- Designed and simulated a precision rotary pick-and-place mechanism in SolidWorks, detailing custom gear cam linkages to optimize the automated packaging workflow.
- Engineered the control architecture utilizing an Allen-Bradley CompactLogix L27ERM PLC and a Kinetix 5500 Servo Drive, achieving high-speed, synchronized servo motor control to improve placement accuracy and reduce mechanical wear.

6. PLANT DESIGN & LAYOUT (NEW DIAPER FACILITY) | ***APR 2025- JUN 2025***

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- Optimized spatial layouts for a new manufacturing plant by converting 2D AutoCAD civil plans into comprehensive 3D SketchUp models.
- Generated photorealistic V-Ray and Enscape walkthroughs, successfully pitching the final conceptual design to the CEO and executive board.

7. TECHNICAL TRAINING DEVELOPMENT (ALLEN-BRADLEY MOTION SYSTEM WORKBENCH) | ***SEP 2025***

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- Designed and wired a complete motion control training workbench utilizing an Allen-Bradley CompactLogix PLC, PowerFlex VFD, Kinetix Servo Drive, and Schneider switchgear to accelerate technician upskilling.

8. IOT DIGITAL CONTROL SYSTEM (DIGITAL OIL LAMP) | ***JAN 2026- FEB 2025***

INDEPENDENT PROJECT

- Architected an end-to-end IoT solution enabling global control of physical hardware devices via the MQTT protocol.
- Developed backend control logic using Python and deployed a responsive frontend web interface via Netlify, demonstrating full-stack capabilities.

9. 3D DIGITAL TWIN & TRACKER (SMART RUBIK'S CUBE) | ***DEC 2025- MAR 2026***

INDEPENDENT PROJECT

- Built a web application interfacing directly with physical hardware (GAN smart cube) via the Web Bluetooth API to stream live telemetry data.
- Utilized Three.js to render a synchronized, real-time 3D digital twin of the hardware and integrated a custom solving algorithm utilizing Computer Vision.

10.REAL-TIME DATA ACQUISITION (DAQ) SYSTEM) | ***DEC 2024***

ACADEMIC MINI PROJECTS

- Developed an ATmega328P microcontroller-based DAQ system tailored for the real-time health monitoring of a robotic manipulator.
- Integrated continuous sensor data collection to facilitate predictive maintenance and mechanical troubleshooting.

LANGUAGES

English
Tamil
Sinhala

EDUCATION

Bachelor of Engineering Technology (Hons) in Instrumentation & Automation ***2021-2026***
Faculty of Technology, University of Colombo

GCE Advanced level Engineering Technology ***2018-2020***
CP/N/Cambridge college kotagala

REFERENCES

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